

AI for Workforce

Module 10: Python
Functions & Packages

Recap

Agenda

1. What are Packages and why do we use them?
2. Which are the common packages we use for AI?
3. What are functions in Python?
4. Some Python exercises
5. What are Neural Networks?

What is Python?

- General purpose coding language
- Easy to learn and use
- Free!



Why are we using Python?

- Easy to read
- Huge repository of Libraries to use
- Flexible
- Quick to see results

Do you know of any other Programming Language?

Python Libraries

Why we use

- No need to write from scratch
- Over 100000 libraries to use
- Vital for developing AI solutions
- We will explore NumPy, Pandas and scikit-learn in this module

Do you know of any other Library?



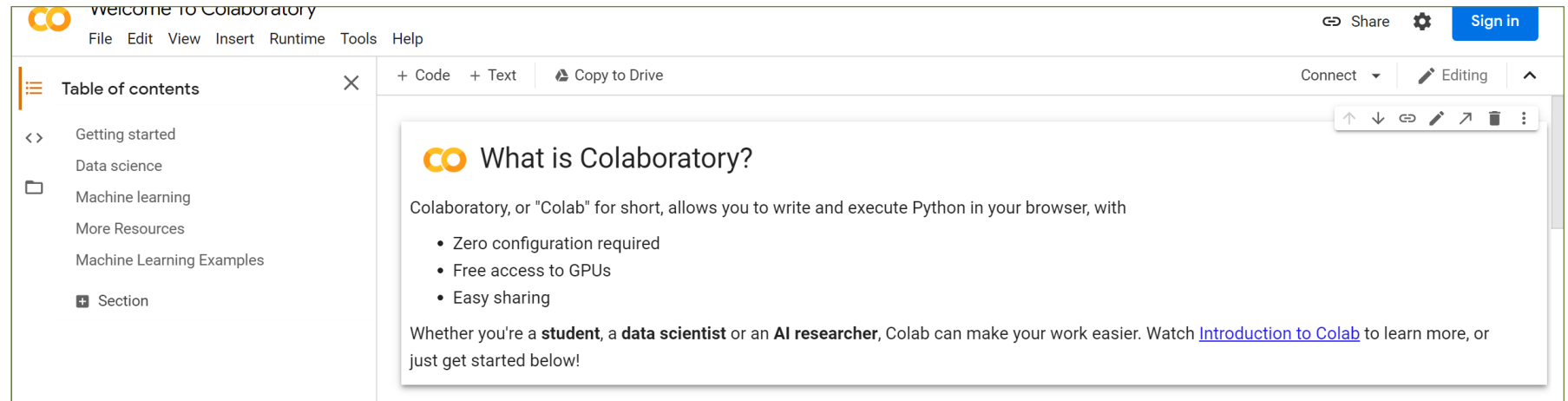
NumPy



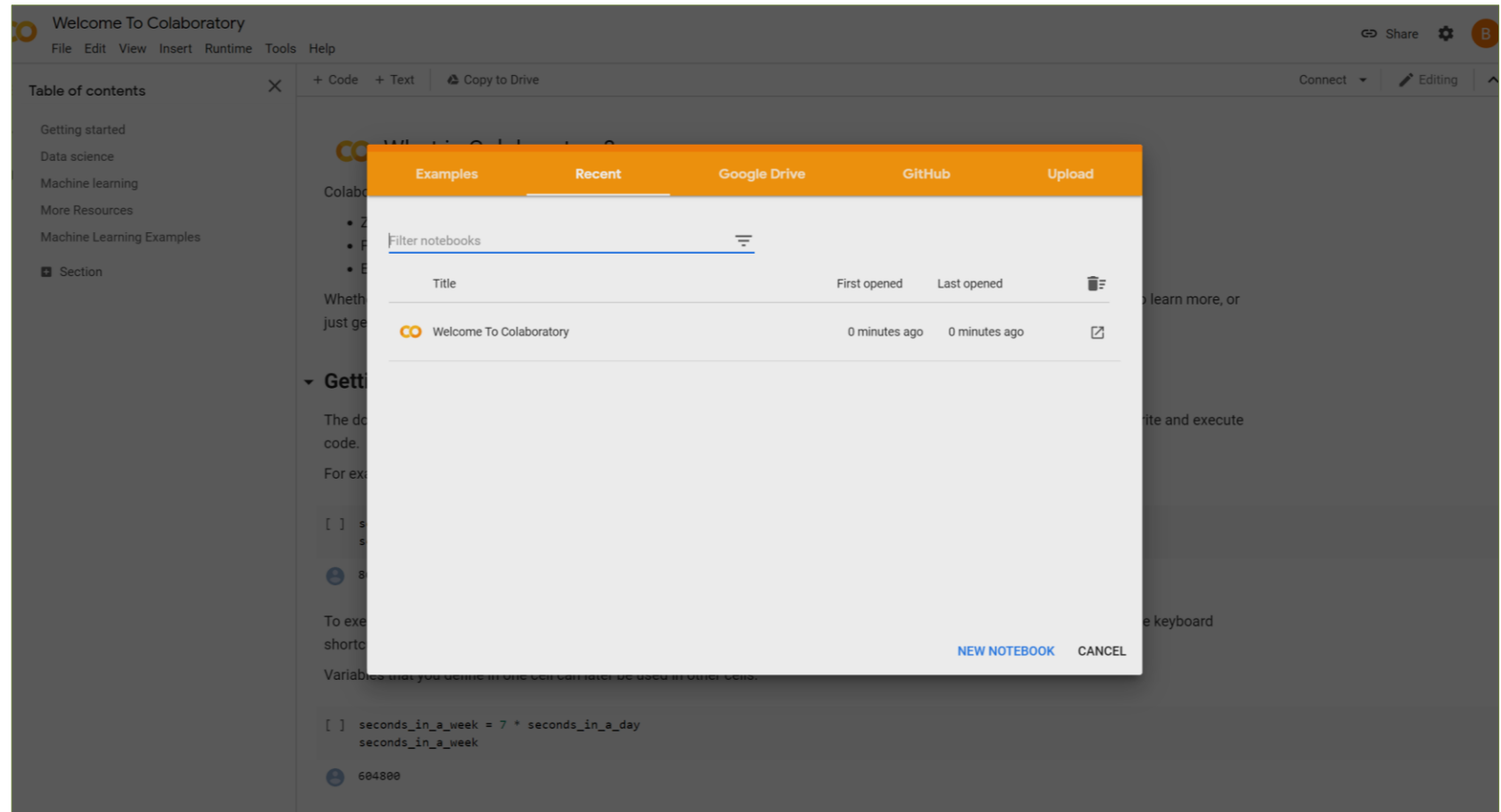
Scikit-learn

Go to Google Colab

Sign In

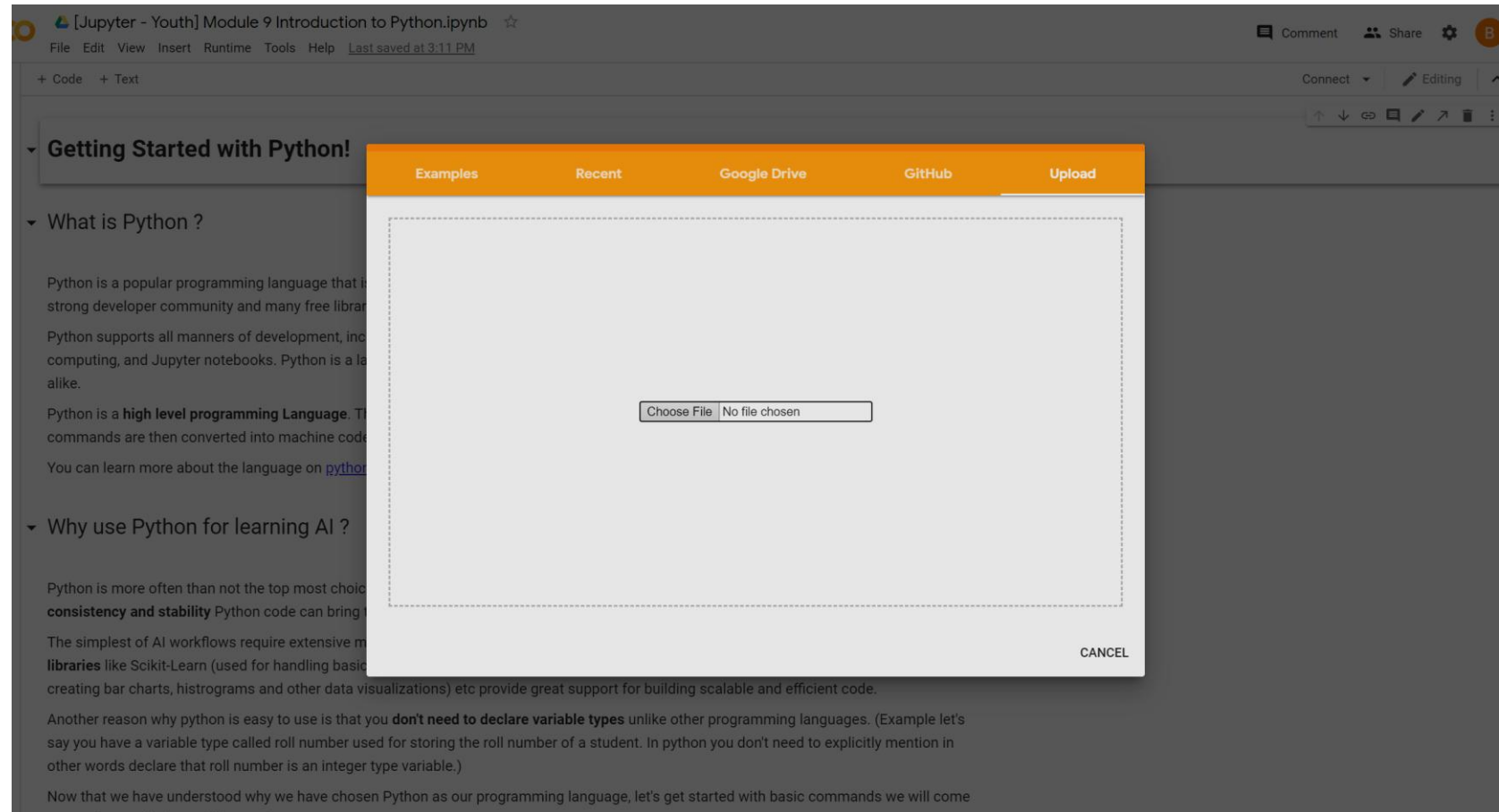


After Logging in



Google Colab. (2020). What is Colaboratory? [Screenshot]. Google Colab.
<https://colab.research.google.com/notebooks/intro.ipynb>

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Google Colab. (2020). Google Colab [Screenshot]. Google Colab. <https://colab.research.google.com/>

Upload File

- Locate [Notebook - student] Module 10 (NumPy & Pandas).ipynb
- Let's Begin!

▼ What is Python ?

Python is a popular programming language that is reliable, flexible, easy to learn, free to use on all operating systems, and supported by both a strong developer community and many free libraries (Libraries are pre-written codes that developers use to solve common programming tasks).

Python supports all manners of development, including web applications, web services, desktop apps, scripting, data science, scientific computing, and Jupyter notebooks. Python is a language used by many universities, scientists, casual developers, and professional developers alike.

Python is a **high level programming Language**. This simply means that the commands used in Python are understandable by humans. These commands are then converted into machine code by using an interpreter.

You can learn more about the language on python.org and [Python for Beginners](#).

▼ Why use Python for learning AI ?

Python is more often than not the top most choice for Machine Learning Applications. The main reason for this is the amount of **simplicity**,

NumPy and Pandas – Let's Begin!

Refer to your Notebook in Google Colab

NumPy Arrays

- Grid of values of the same data type.
- Every element indexed [0,1,2..]
- Shape is size of array

```
[1] #importing numpy package
    import numpy as np
    #initializing an array with the elements given in the brackets
    a = np.array([1,2,3])
    print (a)
```

```
↳ [1 2 3]
```

NumPy array Indexing

- Slicing will always give you a subarray of the original
- Array index allows you to construct a new one by picking and choosing from another array

```
# Example of integer array indexing.  
# Returned array will have shape (3,) and  
b=(a[[0, 1, 2], [0, 1, 0]])  
print(b)  # Prints "[1 4 5]"  
print(b.shape)
```

Datatypes in NumPy Arrays

- NumPy can choose datatype on its own
- Use `dtype=datatype` to assign it yourself



```
x = np.array([1, 2], dtype=np.int64)  
print(x.dtype)
```



```
int64
```

NumPy Array Mathematics

- Basic mathematical operations on an elementwise manner
- Using Operators or Functions built into NumPy itself

Elementwise sum; both produce the array

```
[[ 6.0  8.0]  
 [10.0 12.0]]
```

```
int(x + y)
```

```
int(np.add(x, y))
```

Sorting NumPy Arrays

- Built in functions to sort elements in an array
- Possible to slice and sort the same array

```
np.sort(a) #function for sorting elements in ascending  
print (np.sort(a))
```

```
[1 1 2 2 3 4 5]
```


Manipulating NumPy Arrays

- Reshaping changes the shape
- Elements do not change

```
[ ] y=x.reshape(6,2) #changing shape from 12*1 to 6*2 array  
    print(y)
```



```
[[ 0  1]  
 [ 2  3]  
 [ 4  5]  
 [ 6  7]  
 [ 8  9]  
 [10 11]]
```

Concatenating NumPy Arrays

- Able to concatenate arrays like Strings
- Elements from the first two arrays are joined to the third

```
c = np.concatenate((x,y,z)) #function to perform concatenation  
print(c)
```

```
[ 3 12]  
[12  4  7]  
[2 6 8]  
[ 3 12 12  4  7  2  6  8]
```

Reflection



- Why do you think we use NumPy over Python variables?
- Let's discuss!

Pandas DataFrames

- 2D Array
- Supports many kinds of inputs
- Have functions to read files like `.csv`

```
      date  mean_temperature  ...  min_pressure_2  rainfall
0  2016-05-04             34  ...             1001         0.0
1  2016-05-05             31  ...             1000         0.0
2  2016-05-06             28  ...             1001         5.0
3  2016-05-07             30  ...             1003         0.0
4  2016-05-08             34  ...             1004         0.0
```

```
[5 rows x 17 columns]
```

Pandas DataFrames

- `.head()` function shows first 5 rows of a DataFrame
- `.head()` supports arguments to show more rows

```
➤
```

	date	mean_temperature	...	min_pressure_2	rainfall
0	2016-05-04	34	...	1001	0.0
1	2016-05-05	31	...	1000	0.0
2	2016-05-06	28	...	1001	5.0
3	2016-05-07	30	...	1003	0.0
4	2016-05-08	34	...	1004	0.0
5	2016-05-09	34	...	1002	0.0
6	2016-05-10	34	...	1002	0.3
7	2016-05-11	32	...	1000	0.8
8	2016-05-12	34	...	1002	2.0
9	2016-05-13	34	...	1001	0.3

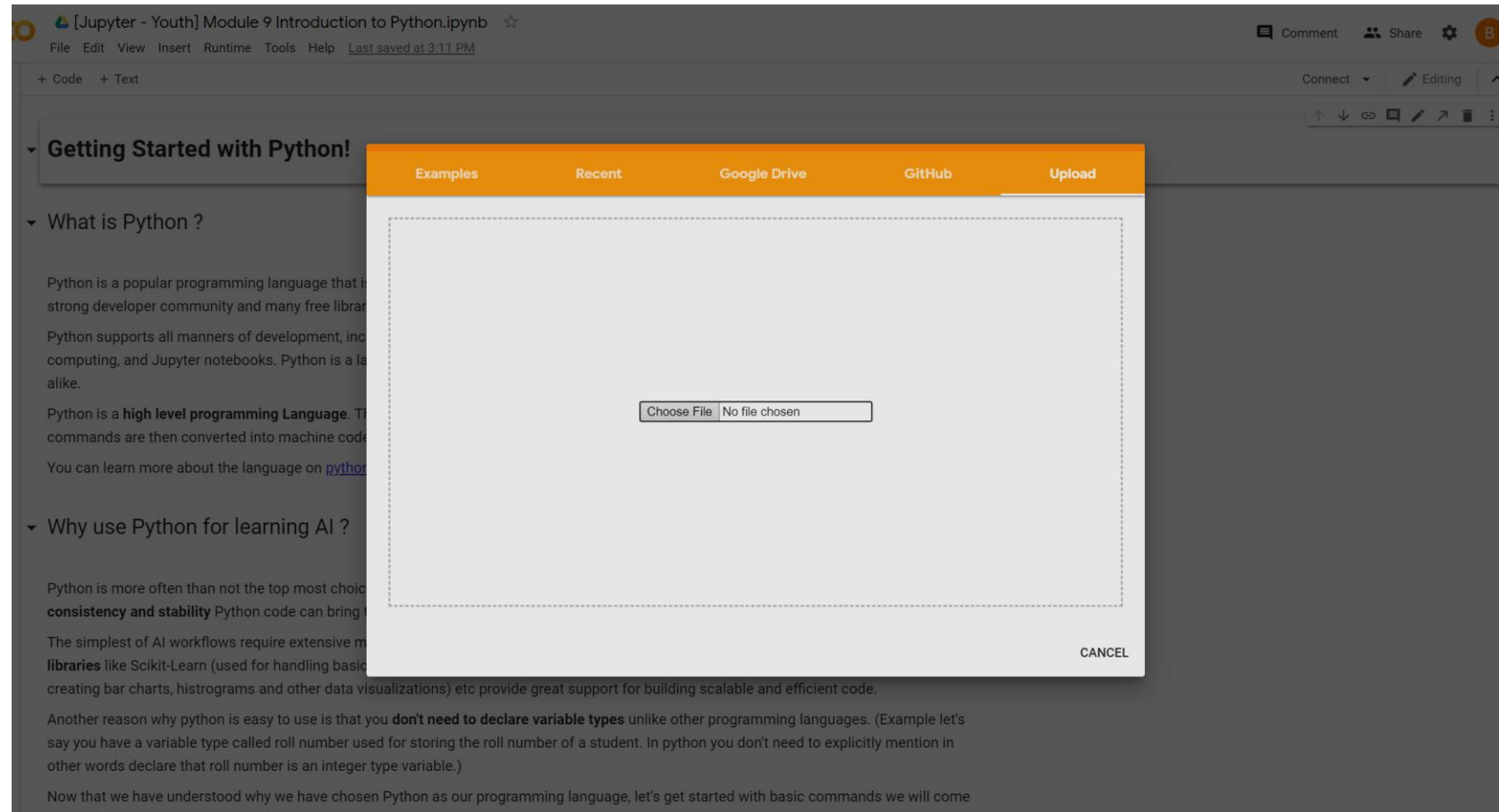
Pandas DataFrames

- Supports many NumPy datatypes
- `.dtypes` shows all the datatypes in the DataFrame

```
dataframe.dtypes
```

date	object
mean_temperature	int64
max_temperature	int64
min_temperature	int64
Mean_dew_pt	int64
mean_pressure	float64
max_humidity	int64
min_humidity	int64
max_dew_pt_1	int64

Go to the Upload tab



Google Colab. (2020). Google Colab [Screenshot]. Google Colab. <https://colab.research.google.com/>

Upload File

- Locate [Notebook - student] Module 10 (matplotlib & Scikit).ipynb
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▼ Why use Python for learning AI ?

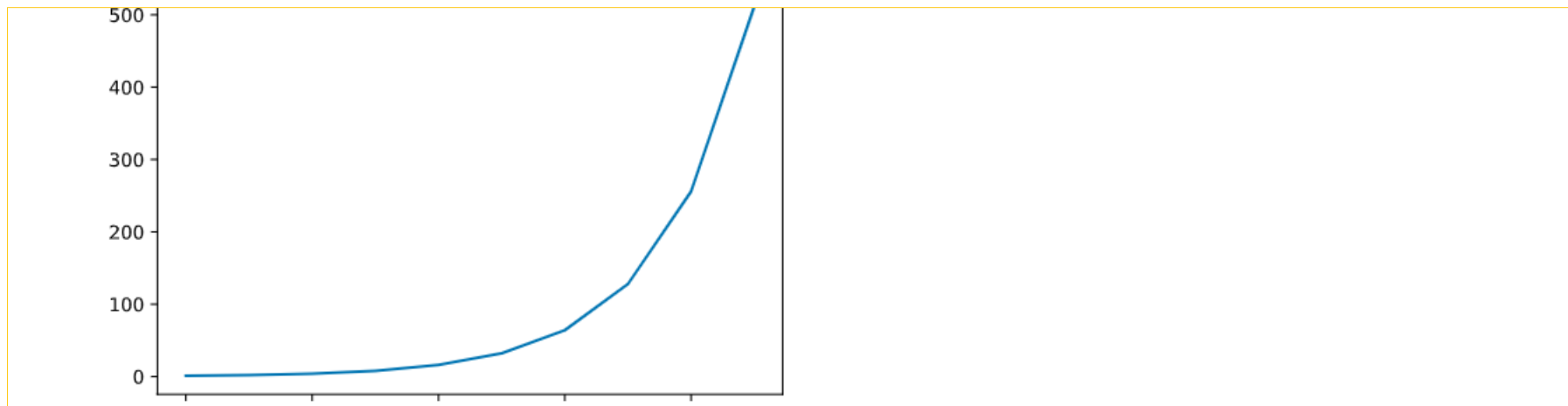
Python is more often than not the top most choice for Machine Learning Applications. The main reason for this is the amount of **simplicity**,

Matplotlib and scikit-learn– Let's Begin!

Refer to your Notebook in Google Colab

matplotlib

- Used to visualize data
- Supports DataFrames and NumPy as well as other libraries



Scikit Learn

- Open Source Machine Learning library
- Various tools provided

```
[ ] #Importing sklearn library
import sklearn
print(sklearn.__version__)
```

Reflection



- What is the relationship between Pandas DataFrames, Scikit Learn and matplotlib and why were they introduced together?
- Let's discuss!

Let's play Kahoot!

[Go to the link](#)

Machine Learning vs Neural Networks

AI for Workforce

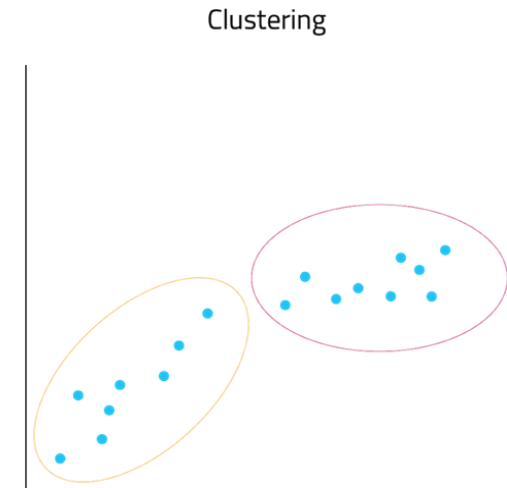
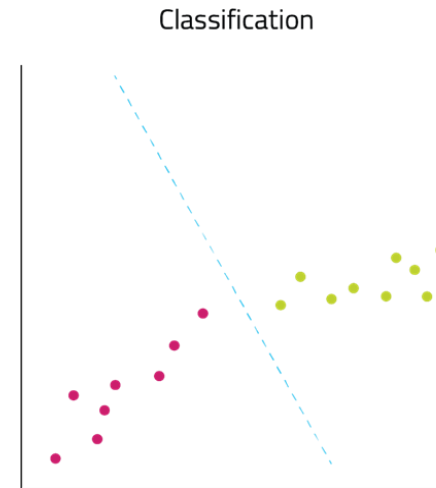
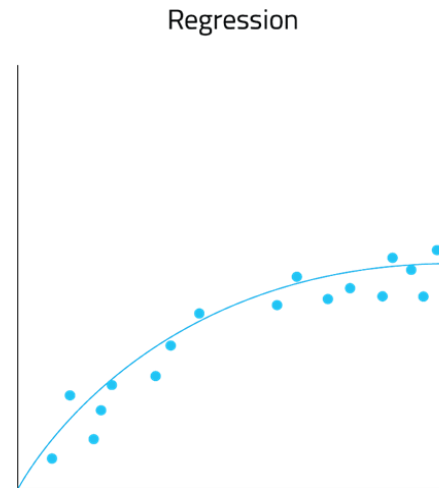
General Approach to AI models

- Rule Based
- Machine Learning Approach



Common Machine Learning Algorithms

- Regression
- Classification
- Clustering

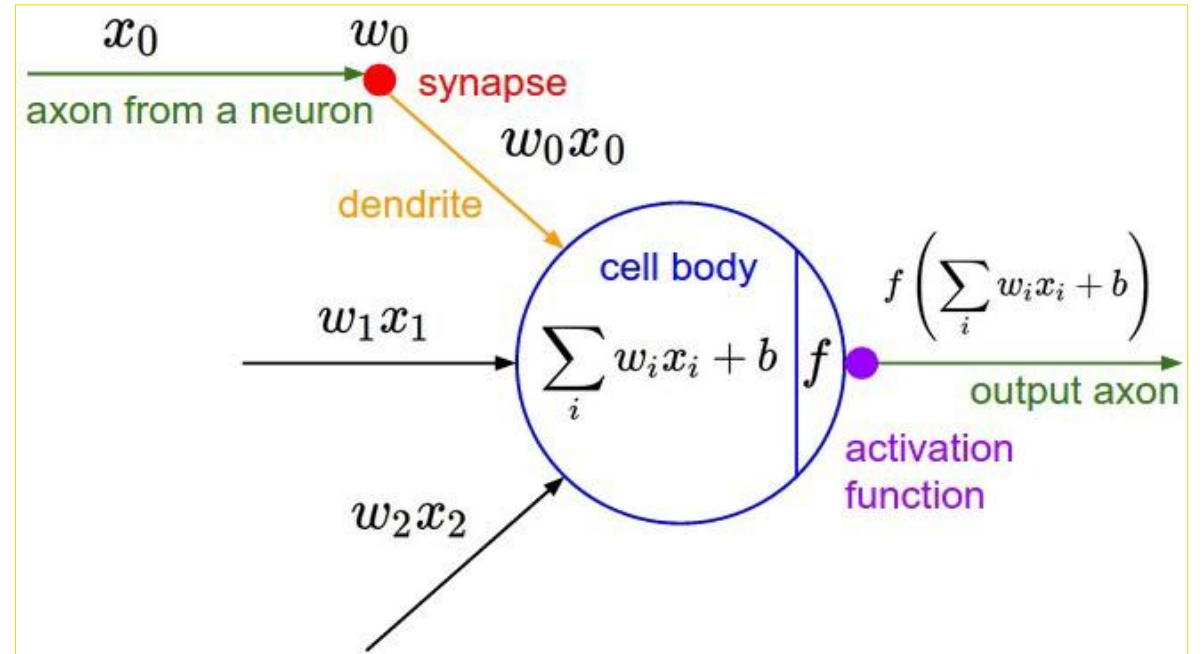
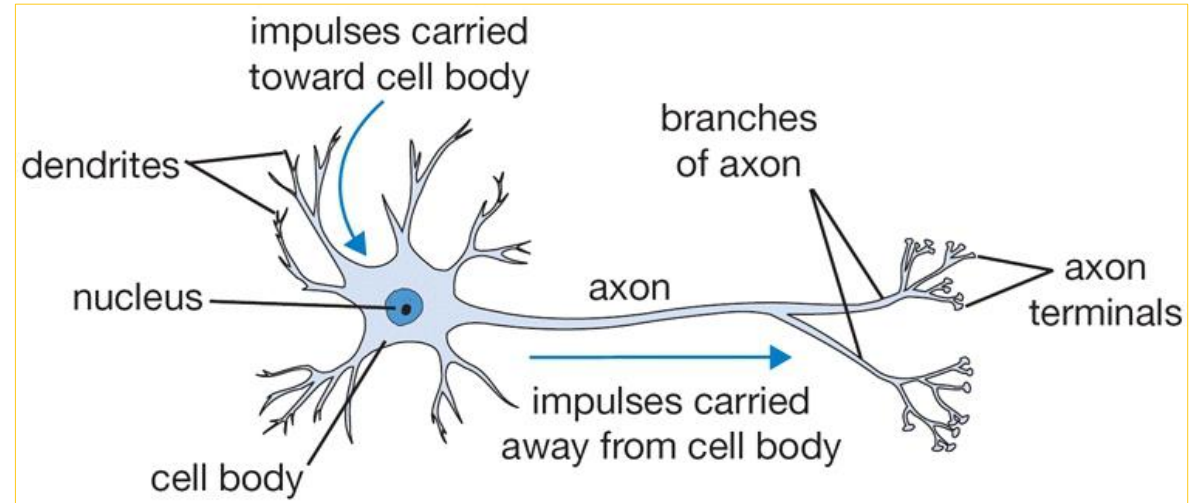


Robert Harper. (2017). Understanding the Machine Learning in AIOps Part 2 – If It Looks Like A Duck... [Image]. Moogsoft
<https://3ijzgg28jrql1xk3g33heonv-wpengine.netdna-ssl.com/wp-content/uploads/2017/08/Regression-Class-Clustering-graph.png>

Neural Network

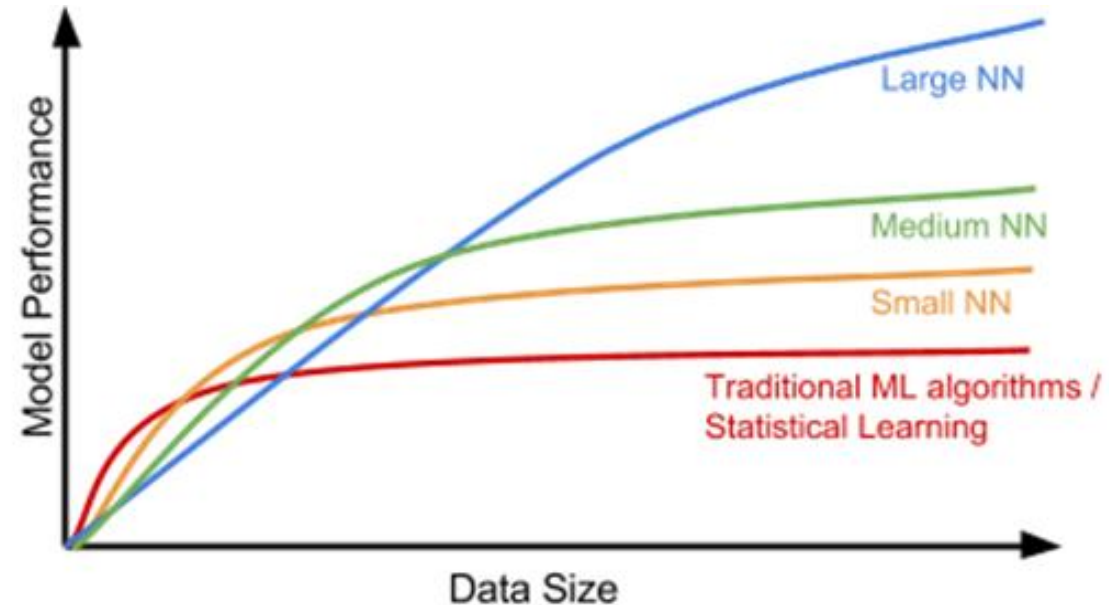
Brain vs Computer

- Neurons carry electrical impulses
- Nodes instead of neurons
- Each node is an ML algorithm



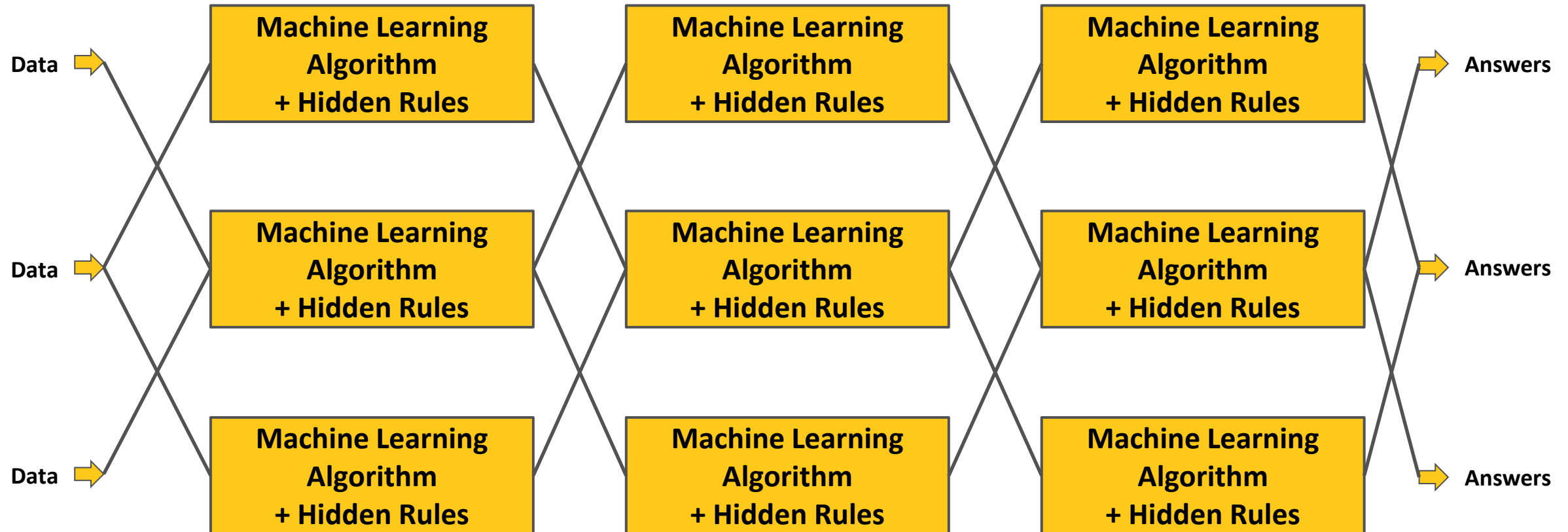
Neural Networks

- Loosely modelled after how neurons work
- Do actions without human input
- Useful for solving problems of large datasets

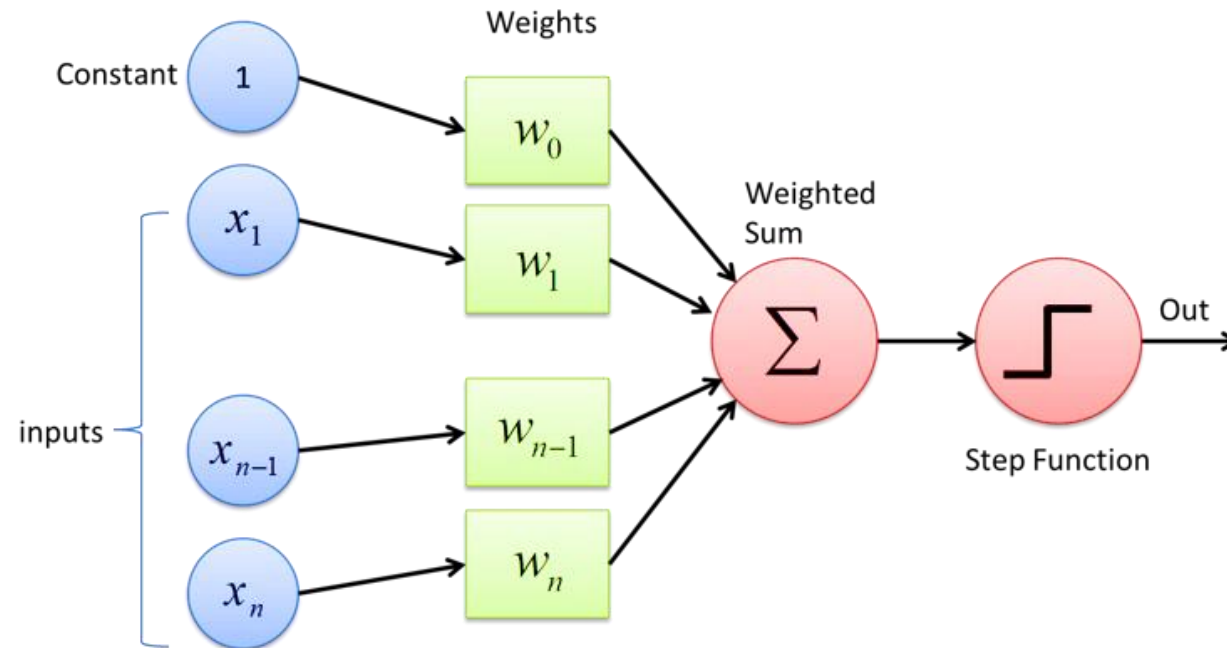


Nvs Abhishek. (2018). Machine Learning: A Gentle Introduction. [Image]. Towards data science
https://miro.medium.com/max/605/1*x6tN3Uqq13BsAMaJvV373w.png

Neural Networks



Do you recall the Perceptron?



[Image of perceptron]. Retrieved October 9, 2020, from <https://deepai.org/machine-learning-glossary-and-terms/perceptron>

Let's watch together

Video

What are Neural Networks and how they work

WHAT ARE NEURAL NETWORKS AND HOW THEY WORK |

today we're going to learn what Neural Networks are and how do they work

Neural Network in code

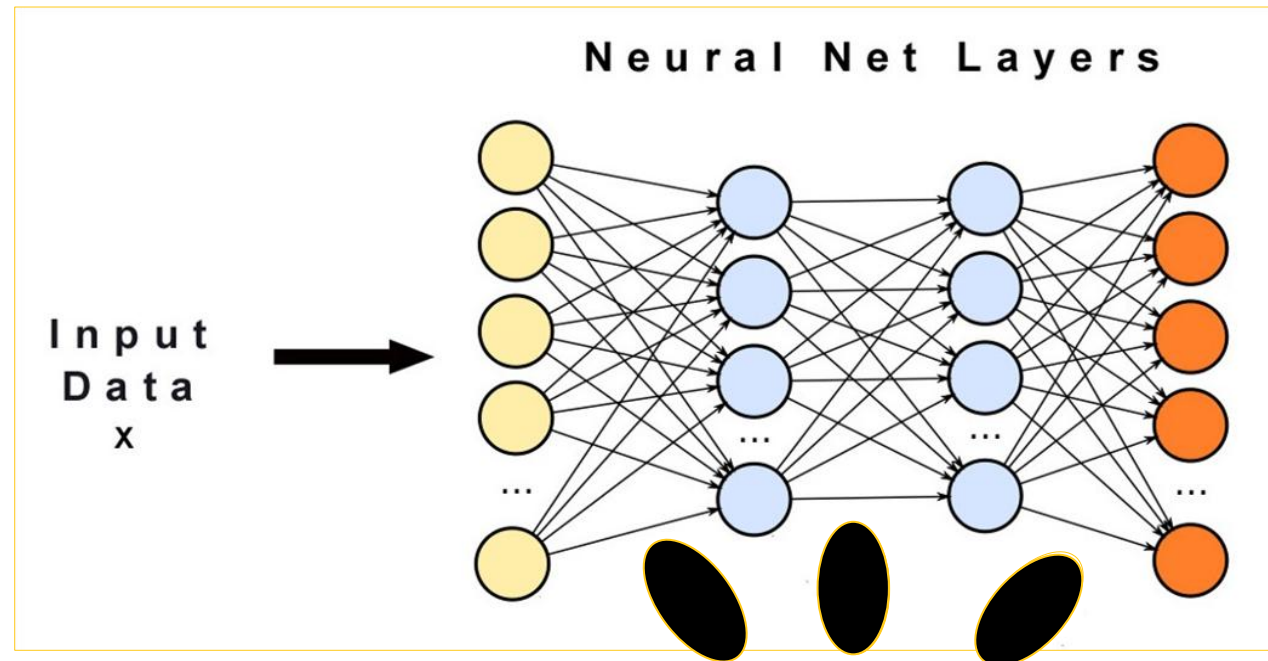
- This is how it looks like in code
- Any keywords you can make out in here?

```
model = Sequential()

model.add(Convolution2D(nb_filters, kernel_size[0], kernel_size[1],
                        border_mode='valid',
                        input_shape=input_shape))
model.add(Activation('relu'))
model.add(Convolution2D(nb_filters, kernel_size[0], kernel_size[1]))
model.add(Activation('relu'))
model.add(MaxPooling2D(pool_size=pool_size))
model.add(Dropout(0.25))
```

Neural Network in code

- This is a visualization of how it works
- Any idea of what each node does?



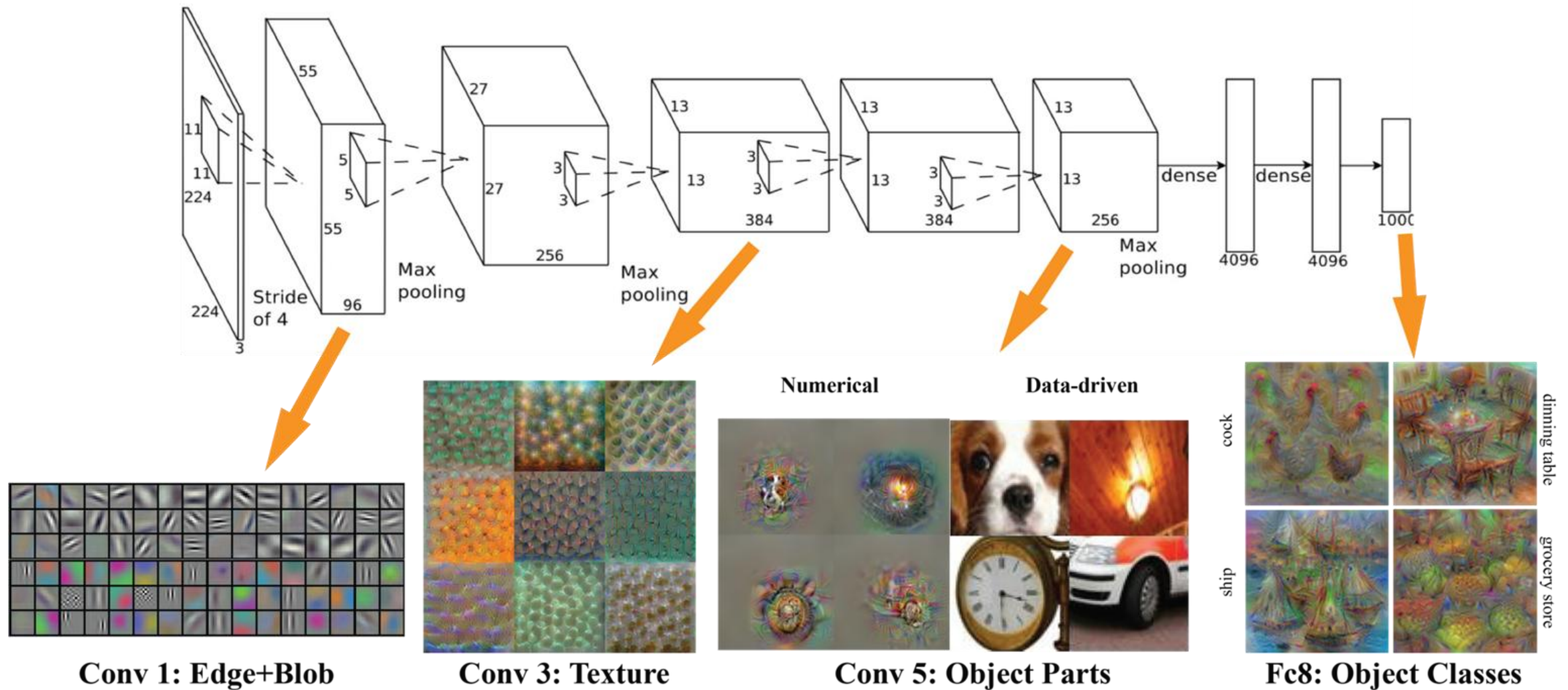
Neural Network in code

- Remember this?
- Take note of the term “Convolution”

```
model = Sequential()

model.add(Convolution2D(nb_filters, kernel_size[0], kernel_size[1],
                        border_mode='valid',
                        input_shape=input_shape))
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model.add(MaxPooling2D(pool_size=pool_size))
model.add(Dropout(0.25))
```


Neural Network in code



Neural Network in code

- What happens?
- Computers see images and pixels in values



What We See

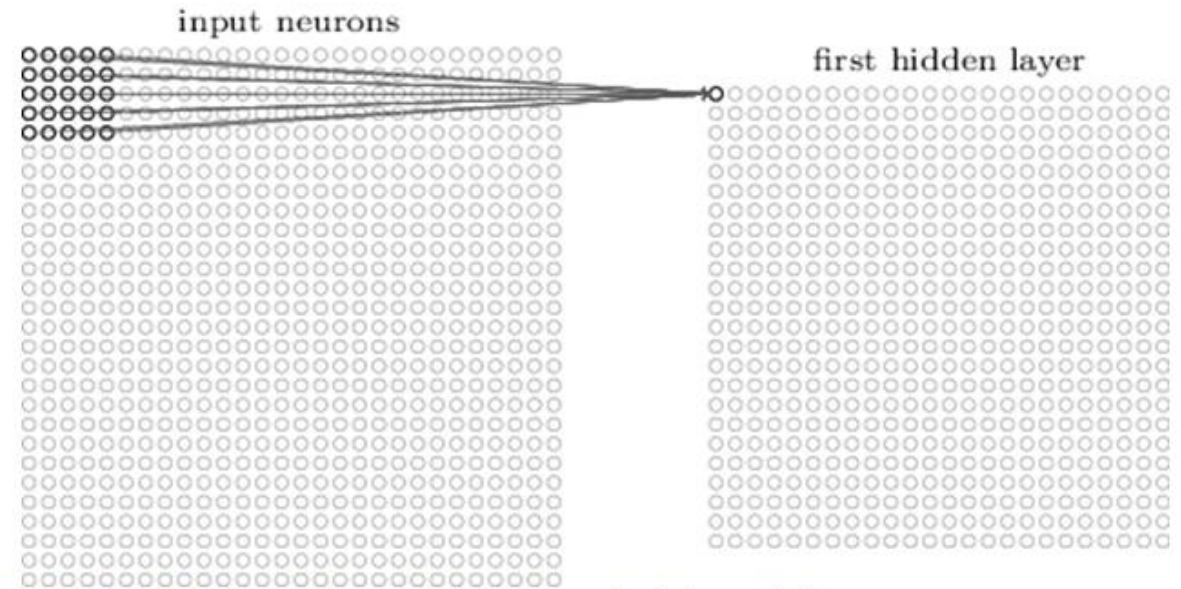
```
08 02 22 97 38 15 00 40 00 75 04 05 07 78 52 12 50 77 31 08
49 49 99 40 17 81 18 57 60 87 17 40 98 43 69 48 04 56 62 00
81 49 31 73 55 79 14 29 93 71 40 67 53 88 30 03 49 13 36 65
52 70 95 23 04 60 11 42 69 24 68 56 01 32 56 71 37 02 36 91
22 31 16 71 51 67 63 89 41 92 36 54 22 40 40 28 66 33 13 80
24 47 32 60 99 03 48 02 44 75 33 53 78 36 84 20 35 17 12 50
32 98 81 28 64 23 67 10 26 38 40 67 59 54 70 66 18 38 64 70
67 26 20 68 02 62 12 20 95 63 94 39 43 08 40 91 66 49 94 21
24 55 58 05 66 73 99 26 97 17 78 78 96 83 14 88 34 89 63 72
21 36 23 09 75 00 76 44 20 45 35 14 00 61 33 97 34 31 33 95
78 17 53 28 22 75 31 67 15 94 03 80 04 62 16 14 09 53 56 92
16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 85 57
86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 58 51 54 17 58
19 80 81 68 05 94 47 69 28 73 92 13 86 52 17 77 04 89 55 40
04 52 08 83 97 35 99 16 07 97 57 32 16 26 26 79 33 27 98 66
88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 57 05 54
01 70 54 71 83 51 54 49 16 92 33 48 61 43 52 01 89 19 67 48
```

What Computers See

Neural Network

What happens?

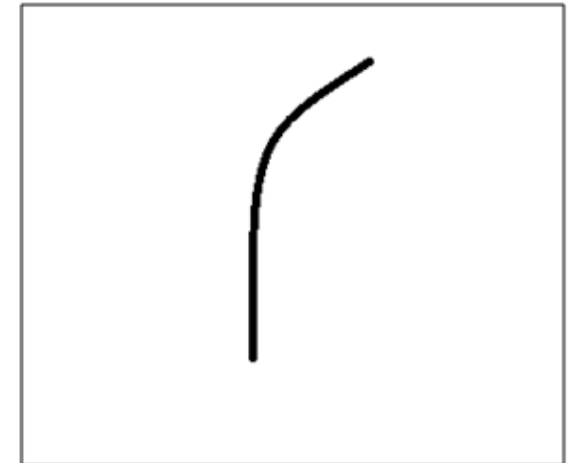
- Input to hidden layers, math matrix operations convolutes the image
- Features like edges and colors are detected



Visualization of 5 x 5 filter convolving around an input volume and producing an activation map

0	0	0	0	0	30	0
0	0	0	0	30	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	30	0	0	0
0	0	0	0	0	0	0

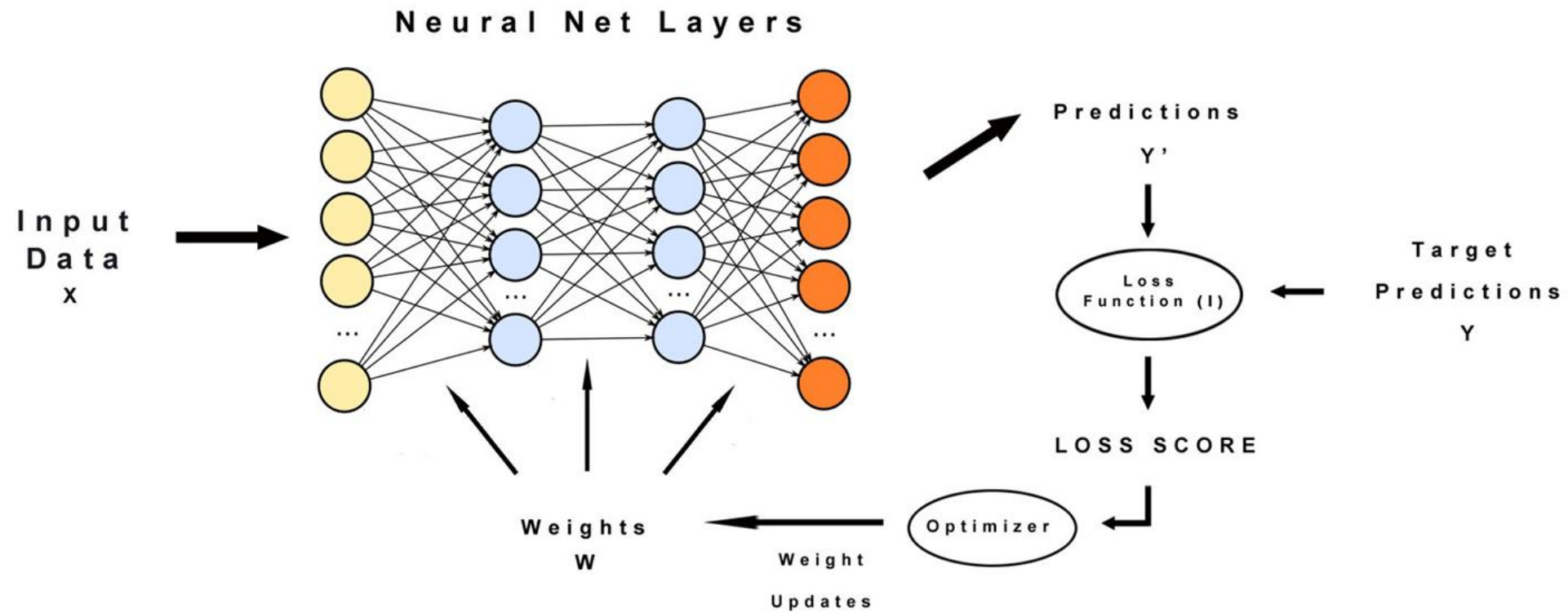
Pixel representation of filter



Visualization of a curve detector filter

Neural Network in code

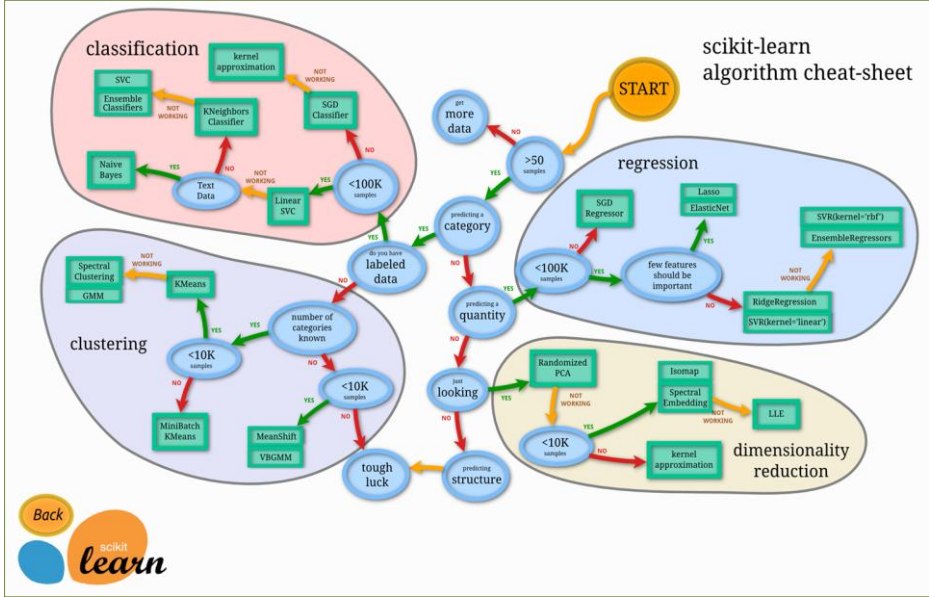
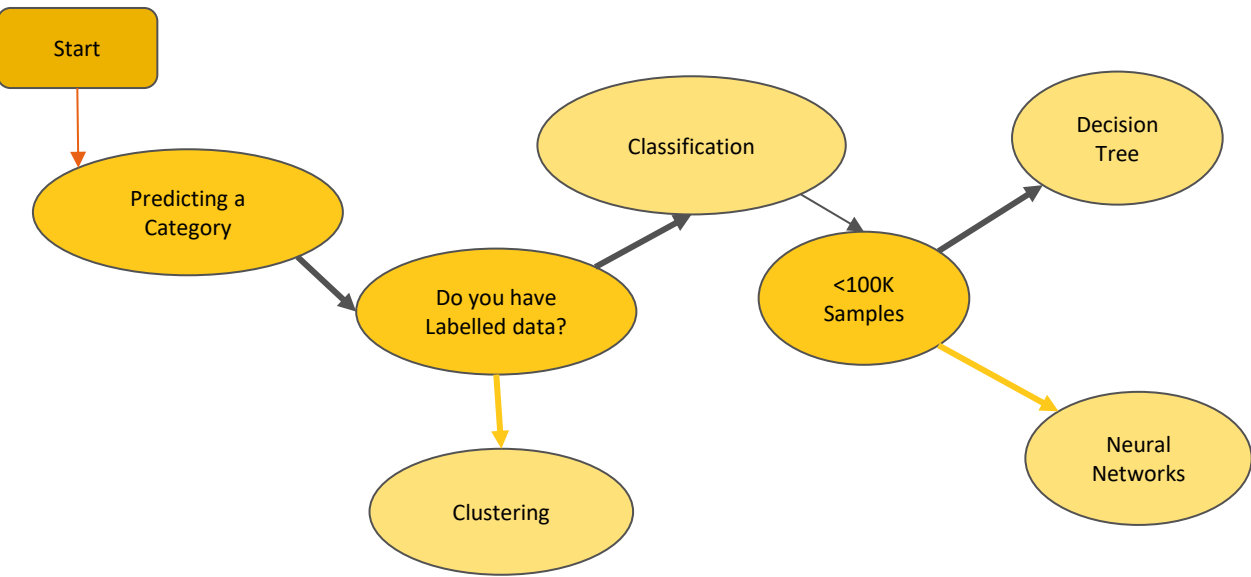
LOGICAL FLOW OF A NEURAL NETWORK



Choosing a Model

When to choose which?

- Use the scikit learn machine learning map



Where do you think Neural Networks are used?

Can you list any application?

Any last words?

In the next module we will be deep diving into AI programming through common trade applications, everything learnt will be put into good use!

Are you ready?

A young man with glasses is shown in profile, looking intently at a computer screen. The background is a blurred classroom with other students. The text 'intel digital readiness' is overlaid on the left side of the image. The Intel logo is in white, and 'digital readiness' is in white with a red-to-orange gradient on the word 'readiness'.

intel. digital readiness



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